

# Milestone 2 — Personal Brief

Mahmoud Hebishy · Student ID 55-18387 · driver-service · Feature **S2-F10 Full-Text Driver Search**  
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## 1. Your assigned tasks

| # | Task                                                                                                                        | Reference      |
|---|-----------------------------------------------------------------------------------------------------------------------------|----------------|
| 1 | <b>S2-F10 Full-Text Driver Search</b> — GET <code>/api/drivers/search/full-text</code> (Elasticsearch + PostgreSQL + Redis) | §10.2.1, p. 36 |
| 2 | §4.3 JWT retrofit on all driver-service M1 endpoints                                                                        | p. 15          |
| 3 | §4.4 Redis caching retrofit on driver-service cached GETs (F1/F3/F5/F6/F9 + CRUD by-id)                                     | p. 15–18       |
| 4 | §7.2 Additive <code>vehicleDetails.description</code> JSONB key on Driver                                                   | p. 28–29       |
| 5 | DP-7 Adapter — <code>ElasticsearchHitAdapter</code>                                                                         | §3.8, p. 13    |
| 6 | CC-6 <code>application.yml</code> for driver-service incl. Elasticsearch fragment                                           | §6.5, p. 26    |

## 2. Start here — tasks that are NOT blocked by anyone

These four can be done independently right now. Land them first so your sub-team (Ahmed Gasser and Ziad Raafat) is unblocked early.

| Task                                                                                         | Why it's unblocked                                                                                                                                             | Suggested order |
|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <b>CC-6 — driver-service <code>application.yml</code></b> (incl. ES, Redis, Mongo fragments) | Pure configuration — only needs the docker-compose containers (CC-5) reachable. Reference fragments are in §6.5, p. 25–26.                                     | 1st             |
| <b>§7.2 Additive <code>vehicleDetails.description</code> key on Driver</b>                   | Purely additive JSONB key on the existing M1 Driver entity — no other M2 work touches it. Existing rows default to empty string.                               | 2nd             |
| <b>§4.4 Redis caching retrofit on driver-service GETs</b> (F1/F3/F5/F6/F9 + CRUD by-id)      | Caching M1 endpoints does not depend on JWT or Elasticsearch. TTLs and key conventions are fully specified in §4.4 and §8.                                     | 3rd             |
| <b>DP-7 Adapter — <code>ElasticsearchHitAdapter</code></b> (skeleton)                        | The interface and basic structure can be written without the final <code>DriverSearchDocument</code> ; only the field mapping needs Ziad's entity to finalize. | 4th (scaffold)  |

### Recommended local-first sequence (concrete commits)

- Branch `feat/cc/CC-6/55-18387` → add `application.yml` with PG, Redis, Mongo, Elasticsearch fragments.
- Branch `feat/m1/MOD-vehicle-description/55-18387` → add the `description` key to `vehicleDetails` JSONB on Driver (default empty string).
- Branch `feat/driver/CACHE-retrofit/55-18387` → wire Redis caching on driver GETs with the documented TTLs and the key convention `driver-service::<entity>::<id>` / `driver-`

`service::<featureId>::<param-hash>`.

- Branch `feat/driver/DP-7/55-18387` → scaffold `ElasticsearchHitAdapter` (interface + skeleton); finalize fields once Ziad's `DriverSearchDocument` lands.

### 3. Blockers ON your work (waiting on teammates)

| Blocker                                                                                                        | Owner                                                | What you need it for                                                                                                                     | Severity    |
|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <code>JwtAuthenticationFilter</code> + <code>SecurityConfig</code> + DP-5 <code>JwtConfigurationManager</code> | <b>Ahmed Gamal Elkot</b><br>(55-24478, user-service) | Required for §4.3 JWT retrofit and CC-1 on every driver endpoint. The spec assigns the reference filter to him; all 5 services reuse it. | <b>HARD</b> |
| <code>DriverSearchDocument</code> <code>Elasticsearch</code> entity (§7.2.1)                                   | <b>Ziad Raafat</b><br>(55-7978, driver-service)      | S2-F10 queries this index document. Scaffold S2-F10 against a stub locally, but the final implementation needs Ziad's entity.            | <b>HARD</b> |

### 4. People you are blocking (downstream of you)

| Blocked teammate                                                   | What they need from you                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ahmed Gasser</b> (55-25085) — S2-F11 Index Driver for Search    | <ul style="list-style-type: none"><li>Additive <code>vehicleDetails.description</code> key on Driver — his indexer reads it.</li><li>Driver-service <code>application.yml</code> with Elasticsearch fragment — his ES writes need the connection.</li><li>Cache-key naming <code>driver-service::S2-F10::*</code> — his §4.4.4 NoSQL-writer invalidation must wildcard-delete it on every ES write.</li></ul> |
| <b>Ziad Raafat</b> (55-7978) — S2-F12 Driver Performance Dashboard | <ul style="list-style-type: none"><li>Your JWT retrofit (his dashboard endpoint is JWT-protected).</li><li>Your Redis caching infrastructure (his dashboard is cached for 10 min).</li><li><code>application.yml</code> for driver-service (Redis and Mongo connections).</li></ul>                                                                                                                           |

### 5. Full M2 task allocation (all 15 members)

#### user-service

| Member                                     | Feature              | Other tasks                                                                                                                                                                                                                                        |
|--------------------------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ahmed Gamal Elkot</b><br>(55-24478)     | S1-F10 Register User | §4.1 BCrypt; reference <code>SecurityConfig</code> + <code>JwtAuthenticationFilter</code> (used by every service); DP-5 Singleton <code>JwtConfigurationManager</code> ; CC-1 JWT on all endpoints; CC-6 user-service <code>application.yml</code> |
| <b>Abdelrahman Mohamed Wael</b> (55-26445) | S1-F11 Login         | §4.3 JWT retrofit on user-service; §4.4 Redis caching retrofit; DP-3 Chain of Responsibility ( <code>AuthHandler</code> ); CC-2 Role management <code>PUT /api/users/{id}/role</code> + seed ADMIN                                                 |
| <b>Seif Tarek Ahmed</b><br>(55-3258)       | S1-F12 Activity      | <code>AuthEvent</code> Mongo doc; Observer retrofit on user writes (S1-F2/F4/F7 + CRUD); Builder retrofit on user DTOs; Adapter <code>ObjectArrayDtoAdapter</code>                                                                                 |

|  |      |           |
|--|------|-----------|
|  | Feed | for S1-F3 |
|--|------|-----------|

## driver-service

| Member                                         | Feature                             | Other tasks                                                                                                                                                                                                                                 |
|------------------------------------------------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Mahmoud Hebishy</b> (55-18387) — <i>YOU</i> | S2-F10 Full-Text Driver Search      | §4.3 JWT retrofit; §4.4 Redis caching retrofit; §7.2 description JSONB key; DP-7 ElasticsearchHitAdapter ; CC-6 driver-service application.yml                                                                                              |
| <b>Ahmed Gasser</b> (55-25085)                 | S2-F11 Index Driver for Search      | Driver CRUD auto-index to ES; §4.4.4 NoSQL-writer invalidation of driver-service::S2-F10::* ; Observer retrofit on driver writes (S2-F2/F4/F7/F8 + CRUD); Builder retrofit on driver DTOs; DP-2 Observer (driver-service MongoEventLogger ) |
| <b>Ziad Raafat</b> (55-7978)                   | S2-F12 Driver Performance Dashboard | DriverEvent Mongo doc; DriverSearchDocument ES entity; DP-4 Builder ( DriverDashboardDTO ); DP-6 Factory DRIVER branch; DASHBOARD_VIEWED logged on every call (incl. cache hits)                                                            |

## ride-service

| Member                                  | Feature                                  | Other tasks                                                                                                                                                                                                          |
|-----------------------------------------|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Mohamed Khaled Naguib</b> (55-25378) | S3-F10 Ride Analytics Dashboard          | §4.3 JWT + §4.4 Redis caching retrofit on ride-service; DP-4 Builder ( RideAnalyticsDashboardDTO ); ANALYTICS_VIEWED logged outside cache; CC-6 ride-service application.yml                                         |
| <b>Ahmed Wael Ibrahim</b> (55-13512)    | S3-F11 Record User-Driver Riding Pattern | Observer retrofit on ride writes (S3-F2/F4/F7/F8 + CRUD); Builder retrofit on ride DTOs; §4.4.4 NoSQL-writer invalidation of ride-service::S3-F12::* ; DP-2 Observer (ride-service); DP-7 Adapter Neo4jRecordAdapter |
| <b>Youssef Mohamed Malek</b> (55-24816) | S3-F12 Driver Recommendations            | RideEvent Mongo doc; Neo4j entities ( UserNode , DriverNode , RODE_WITH ); DP-6 Factory RIDE branch; ownership check via JWT uid                                                                                     |

## location-service

| Member                              | Feature                             | Other tasks                                                                                                                                                                                          |
|-------------------------------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Omar Elharridy</b> (55-0654)     | S4-F10 Location Analytics Dashboard | §4.3 JWT + §4.4 Redis caching retrofit on location-service; DP-4 Builder ( LocationAnalyticsDTO ); ANALYTICS_VIEWED outside cache; Builder retrofit on location DTOs                                 |
| <b>Ahmed El-Mosallamy</b> (55-0823) | S4-F11 Record Location GPS Event    | Observer retrofit on location writes (S4-F2/F4/F7 + CRUD); §4.4.4 invalidation location-service::S4-F12::{driverId} + S4-F10::* ; DP-2 Observer (location-service); DP-7 Adapter CassandraRowAdapter |
| <b>Youssef Maged</b> (55-2829)      | S4-F12 Location Tracking Timeline   | LocationEvent Mongo doc; Cassandra LocationTrackingEvent entity + keyspace bootstrap; DP-6 Factory LOCATION branch; CC-5 docker-compose with all 6 DBs; CC-6 location-service application.yml        |

## payment-service

| Member                                  | Feature                                                  | Other tasks                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Yahia Hesham</b><br>(55-25376)       | S5-F10 Fare Revenue by Vehicle Type with Surge Breakdown | §4.3 JWT + §4.4 Redis caching retrofit on payment-service; Builder retrofit on payment DTOs; DP-4 Builder ( <code>VehicleTypeRevenueDTO</code> ); CC-6 payment-service <code>application.yml</code>                                                                                                                           |
| <b>Seifeldin Hesham</b><br>(55-0664)    | S5-F11 Payment Method Breakdown                          | <code>PaymentAuditEvent</code> Mongo doc (incl. method/amount required-fields contract); Observer retrofit on payment writes (S5-F5/F7 + CRUD); §4.4.4 Observer-driven invalidation for S5-F10/F11; DP-2 Observer (payment-service); DP-6 Factory <code>PAYMENT_AUDIT</code> ; DP-7 Adapter <code>MongoDocumentAdapter</code> |
| <b>Seif Tarek Mostafa</b><br>(55-24853) | S5-F12 Process Ride Refund with Surge Handling           | §4.6 add <code>surgeFee</code> key to <code>Payment.transactionDetails</code> (update M1 S5-F4); update M1 S5-F4 to accept <code>?simulateFailure=true</code> ; update M1 S5-F2 to emit <code>REFUNDED</code> audit event; DP-1 Strategy ( <code>RefundStrategy</code> + 3 strategies + selector)                             |

## 6. Recommended sequence for the sub-team

1. Coordinate with **Ahmed Gamal Elkot** on the JWT filter — without it, your CC-1 work is blocked (and so is everyone else's).
2. Land your **§7.2 description key** and **driver-service application.yml** early so Ahmed Gasser and Ziad can move.
3. Sync with **Ziad Raafat** on the `DriverSearchDocument` field shape so you can scaffold the S2-F10 query in parallel.
4. Define the `driver-service::S2-F10::*` cache-key naming and confirm it with Gasser, so his §4.4.4 invalidation matches your keys.